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EXPERIMENT - 8

SEMPHORE

AIM:

To write and execute a C program for Inter Process Communication using Semaphore.

CODE:

```
#include<stdio.h>
#include<stdlib.h>
int mutex=1,full=0,empty=5,x=0;
int wait(int s){
    return --s;
}
int signal(int s){
    return ++s;
}
void producer(){
    mutex=wait(mutex);
    empty=wait(empty);
    full=signal(full);
    x++;
    printf("\nProducer produces item %d",x);
    mutex=signal(mutex);
}
void consumer(){
    mutex=wait(mutex);
    full=wait(full);
    empty=signal(empty);
    printf("\n consumer consumes item %d",x);
    x--;
```

```
int main(){
    int n;
    printf("\n producer consumer problem");
    printf("1.producer");
    printf("2.consumer");
    printf("\n 3.EXIT");
    while(1){
        printf("\nEnter your choice: ");
        scanf("%d",&n);
        switch(n){
            case 1:
                if(mutex==1 && empty!=0)
                {
                    producer();
                }
                else{
                    printf("\n Buffer is Full!! Producer cannot produce.");
                }
                break;
            case 2:
                if(mutex==1 && full!=0){
                    consumer();
                }
        }
    }
}
```

```

    }
    else{
        printf("\n Buffer is Full!! Producer cannot produce.");
    }
    break;
case 2:
    if(mutex==1 && full!=0){
        consumer();
    }
    else{
        printf("\n Buffer is empty!! Consumer cannot consume.");
    }
    break;
case 3:
    printf("\nExiting program...");
    printf("\nFinal buffer items producer : %d\n",x);
    exit(0);
default:
    printf("\nInvalid choice .try again");
}
}
return 0;

```

OUTPUT:

```

user@123@localhost:~ — ./a.out
[user@123@localhost ~]$ gcc sender.c
[user@123@localhost ~]$ ./a.out
Enter message: hello
Data written to shared memory: hello
[user@123@localhost ~]$ vi sender.c
[user@123@localhost ~]$ vi receiver.c
[user@123@localhost ~]$ vi semaphores.c
[user@123@localhost ~]$ gcc semaphores.c
[user@123@localhost ~]$ ./a.out

producer consumer problem
1.producer
2.consumer
3.Exit
Enter your choice:

```

RESULT:

Thus,the Inter Process Communication using Semaphore was implemented successfully and the expected output was obtained.